Operational Amplifier

All question from 5 session

2018-19

3) a. Show the pin ~~diagram~~ configuration of a typical op-amp with it's function? - 4

b. Draw the circuit of an op-amp as voltage follower and find an expression for it's voltage gain? - 6

c. Draw the circuit for a non inverting amplifier and explain how it works? - 4

4) a. Show the Differences between open loop gain and close loop gain? - 3

b.) Show the circuit diagram of integrator and differentiator and explain how it works - 6

c. Draw the circuit diagram of substractor

P.T.O

and explain how it works? – 5 Marks

17-18

3) a. Show the circuit of symbol and terminals of operational amplifier 741C. – 4 Marks

b. Show block diagram for using op-amp in sensor based data acquisition – 4 Marks

c. Draw the circuit for an inverting amplifier. Deduce the equation of output voltage, current and close loop gain for inverting amplifier? – 6 Marks

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4) a. Explain how an operational amplifier can be used to design voltage follower? – 4 Marks

b. Show the circuit diagram of integrator and differentiator and explain how it works? – 6 Marks

c. ~~What is quantization error? How it is used~~ How an operational amplifier can be used to build inverting adder? – 4 Marks

16-17 Session

3. b. What are differences between open loop gain and close loop gain of operation amplifier? – 3 Marks

c. Show how can an operational amplifier be used as an adder and subtractor? – 6 Marks

9) a. Explain how an operational amplifier be used as an integrator? – 5 Marks

5(b) → b. Explain how an op-amp can be used as voltage follower? – 4 Marks

9(b) → b. Show how a low pass filter is designed using the operational amplifier? – 5 Marks

[illegible]

c. How voltage gain is obtained for inverting amplifier? – 4 Marks

5) a. Show the circuit diagram of a multichannel amplifier. Show the output voltage analysis and circuit diagram of differentiator? - 6 Marks

b. Explain how op-amp can be used as voltage follower? - 9 Marks

c. Give the examples where D/A and A/D converters are used? - 9 Marks

Mix from AD/DA

2015-16 Session

3. a. Write down the characteristics of ideal op-amp? - 3 Marks

b. What is open loop gain and close loop gain? Show the pin configuration of a typical op-amp? - 5 Marks

c. How output voltages are calculated for

P.T.O

the circuit of i) non inverting amplifier
ii) summing " -6 Marks

4) a. How cascaded op-amp circuit is designed? How output voltage is obtained in cascaded-op amp circuit? - 7 Marks

b. Design an op-amp circuit with inputs v_1 and v_2 such that $v_o = -7v_1 + 3v_2$ -4 Marks

c. How voltage gain is obtained for inverting amp? - 3 Marks

5.) a) Show the output voltage analysis and circuit diagram for differentiator? -5 Marks

b. Explain why capacitor and inductor are very useful in electronic circuit? - 9 Marks
Mix maybe!!!

c. Design an analog computer circuit to solve the differential equation?

PTO

$$\frac{d^2}{dt^2}v_o + 3\frac{d}{dt}v_o + 2v_o = 4\cos 10t \quad t>0$$

subject to $v_o(0) = ?$, $v_o'(0) = 0$ → Marks 5

একেবারে new type question

2011-12 session

No question from op-amp!!!

2012-13 session

9) a. What is an operational amplifier? What are the fundamental differences between ideal and practical op-amp (Model 741C)
 – 9 Marks

new type. ← b. Solve the differential equation using appropriate circuit diagram?

$D^3y + 4D^2y + 6Dy + 8y = -5$ volts where $D = \frac{d}{dt}$

– 4 Marks

c. write short notes on i) op-amp as a substractor
ii) Frequency scaling
iii) Schmitt trigger.
-6 Marks

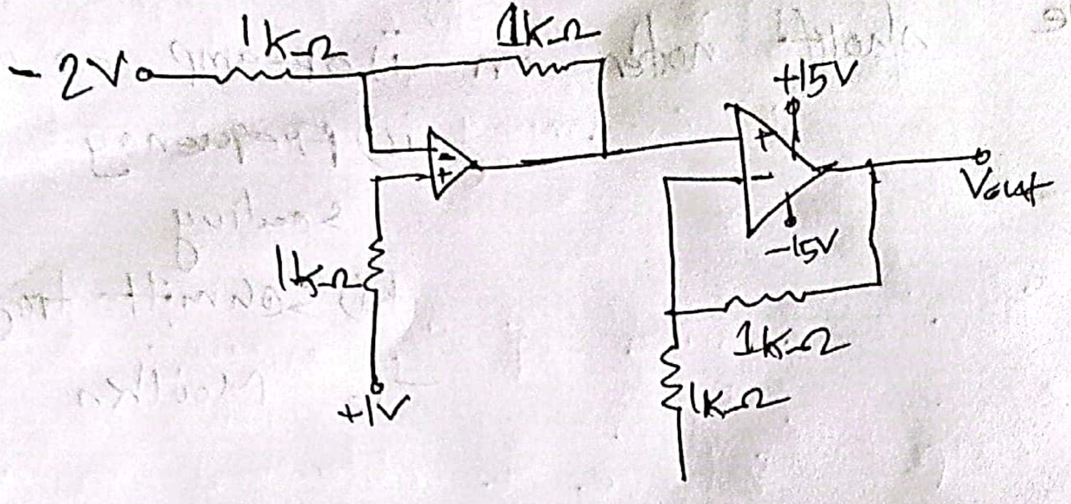
13-19 session

4.a. what is an operational amplifier?
"The inverting amplifier provides a constant voltage gain with 180 phase shift" - Justify the statement with circuit diagram → 5 Marks

b. Define ~~#~~ slew rate and active device solve the differential equation using app... cir... dia...

$D^3y - 9D^2y + 6Dy - 8y = -f(t)$ volts where $D = \frac{d}{dt}$ - 5 Marks

c. calculate the output voltage (Vout) for the circuit of fig 4(c)



5(b) Sketch the equivalent ckt of ideal op-amp. Suppose for a non inverting amp, the equivalent circuit of op-amp is shown in fig 5(b).

i) Find Vs / Vout for Rin = 100kΩ

ii) Vs / Vout for an ideal op-amp